PART A - THEORETICAL FOUNDATION

1. What is Statistical Distribution?

A statistical distribution shows how data values are spread across different ranges. It helps us understand patterns like where most values lie and how data behaves. Common examples include normal, binomial, and Poisson distributions. It is useful for analysis and prediction.

2. What is a Q-Q Plot and why is it used?

A Q-Q (Quantile-Quantile) plot is a graph used to check if data follows a normal distribution. It compares actual data with expected normal values. If points lie on a straight line, the data is normal. It is mainly used for checking data assumptions.

3. Difference between Discrete and Continuous Distributions

Discrete distributions deal with countable values like number of transactions (0,1,2...). Continuous distributions deal with measurable values like height, weight, or amount. Discrete data is separate, while continuous data can take any value within a range.

4. What is Bernoulli Distribution?

Bernoulli distribution is the simplest distribution with only two outcomes: success or failure. It is used when an experiment happens once. For example, transaction success (1) or failure (0). It is the base for binomial distribution.

5. What is Binomial Distribution?

Binomial distribution shows the probability of a number of successes in multiple trials. Each trial has only two outcomes (success/failure). It depends on number of trials (n) and probability (p). Example: number of successful transactions in a week.

6. Explain Log-Normal Distribution

A log-normal distribution occurs when the logarithm of data is normally distributed. It is usually right-skewed, meaning most values are small and few are very large. It is common in income, sales, and transaction data. It helps model real-world financial data.

7. Explain Power Law Distribution

Power law distribution shows that a small number of values are very large while most are small. It follows the “few big, many small” pattern. It is often seen in wealth distribution or online sales. It is also called heavy-tail distribution.

8. What is Box-Cox Transform?

Box-Cox transformation is used to make data more normal (less skewed). It stabilizes variance and improves analysis. It applies a mathematical transformation based on a parameter lambda (λ). It is useful before applying statistical models.

9. Explain Poisson Distribution with an example

Poisson distribution shows the number of events occurring in a fixed time period. It is used when events happen randomly but at a constant rate. Example: number of transactions per day. It depends on average rate (λ).

10. What is Z-score Probability?

Z-score measures how far a value is from the mean in terms of standard deviation. It helps identify outliers and unusual values. Using Z-score, we can calculate probability of values above or below a point. It is widely used in statistics.

11. Difference between PDF and CDF

PDF (Probability Density Function) shows the likelihood of a value occurring. CDF (Cumulative Distribution Function) shows the total probability up to a certain value. PDF gives exact density, while CDF gives cumulative probability. Both help understand data distribution.